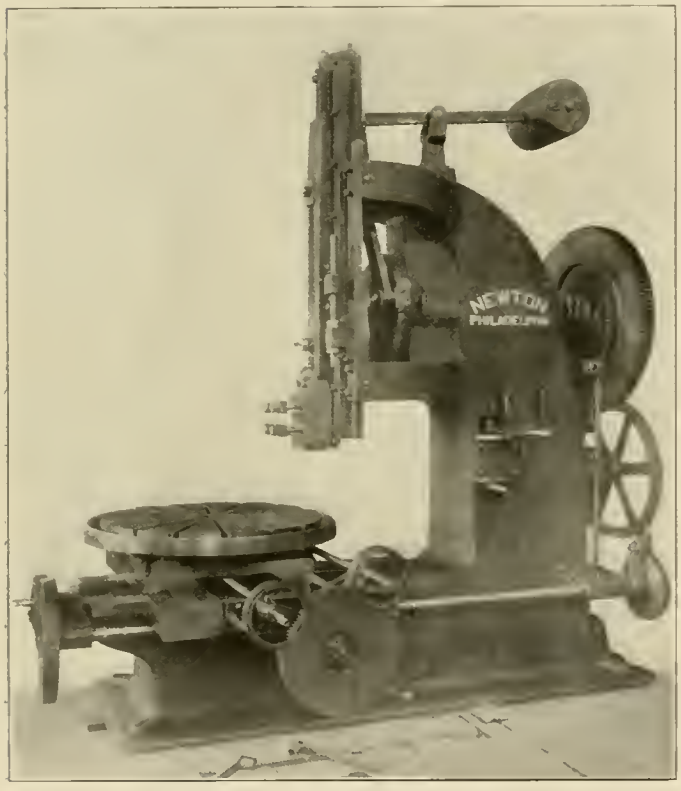
sliding sleeves on which the several groups of gears are mounted. The arrangement is such that the lower train of gears revolves continuously in an oil bath, thus insuring efficient lubrication.

The machine is driven by a 10 hp. motor which is attached to one side of the upright, motion being transmitted through spur gears to the large plate gear shown at the back of the machine and thence to the cutter bar through a crank disc arranged to give a quick return stroke. All dangerously exposed gears are covered with meshed guards which are not shown in the illustration.

The cutter bar, which is of heavy box construction, carries a tool apron designed to relieve the tool or to be fixed as desired, and which has vertical and horizontal steel faced clamping surfaces. The bar has square lock bearings in the guide and a continuous taper shoe for side adjustment. The vertical adjusting screw is relieved of strain when cutting by means of a serrated clamping surface on the front face of the bar. The adjusting screw is operated by means of a square shaft projecting from the face of the bar. The cutter bar guide has square lock bearings



Slotting Machine With Inside Speed Box

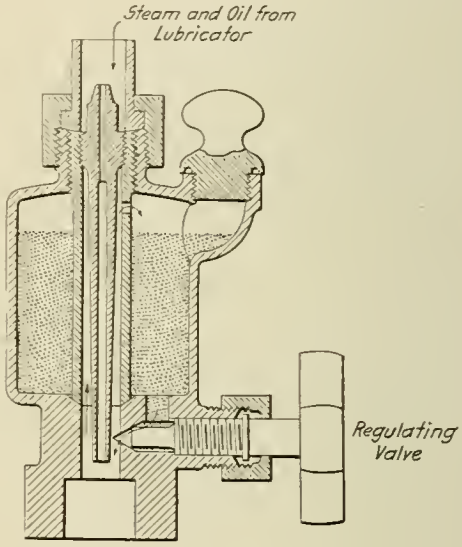
in the frame and is adjusted vertically to suit the location of the bar by means of a hand screw shown in the illustration. The cutter bar yoke has a bearing on each side of the connecting rod thus supporting the pin in double shear, and the stroke adjusting screw is relieved of strain by tongues on the connecting rod washer which engage grooves in the face of the crank pin.

The work table is surrounded by an oil pan forming an integral part of the casting, the periphery of which is graduated in degrees. The horizontal adjustment of the saddle on the cross slide and of the cross slide on the base are made by means of taper shoes. Power feed connections are provided for all motions of the table, including the circular motion of the table itself, as well as the in-and-out and cross movement of the saddle and cross slide respectively. For ordinary requirements the machine is designed to give a range of cutting strokes varying from 8 to 40 strokes per minute, but where desired the limits of the range may be altered, the ratio, however, remaining the same.

GRAPHITE LUBRICATOR

The Graphite Lubricator Company, Detroit, Mich., has placed on the market a graphite lubricator for lubricating with flake graphite the valves and cylinders of locomotives. This graphite is suspended in common engine oil for the purpose of more easily filling the lubricator. The device is used in connection with the oil lubricator as now in use on locomotives. The present oil lubricator acts as the cleaner, while the graphite lubricator delivers graphite to the cylinders and valves as the lubricant. The accompanying illustration clearly shows its construction. The oil line from the steam or gravity-feed lubricator in the cab is connected to the top of the graphite lubricator, and the oil and steam pass to the cylinder and valves through the nozzle shown on the inside. The graphite is contained in the annular space surrounding this nozzle, and the feed is controlled by means of the regulating valve, as shown. The pulsating pressure of the steam will cause the feeding of the graphite through the passage controlled by the regulating valve, and there it becomes mixed with the oil from the gravity-feed lubricator. It is then distributed to the various parts of the valves and cylinders.

This lubricator is adapted for use on superheater and compound locomotives (either piston or slide valve type) and especially those engines which experience trouble in obtaining the



Cylinder Lubricator for Flake Graphite

proper lubrication of the valves and cylinders. It is claimed that the priming of locomotives will not in any way interfere with the proper lubrication of the cylinders and valves where this lubricator is used. It is also claimed that 50 per cent less cylinder oil will be required for lubrication in the cylinders and valves, and that at the same time a better lubrication will be obtained by the use of the flake graphite.

The lubricator shown, which will hold three ounces of the graphite lubricating mixture, is filled through the plug shown at the top and to the right of the lubricator. It may be attached to any standard steam chest choke-plug fitting, and thus eliminates the use of the choke plugs. One filling of the model A three-ounce lubricator is sufficient for the operation of a locomotive for approximately 250 miles without further attention on the part of the crew. At present the lubricators are made in two sizes—model A, which holds three ounces of lubricant, and model B, which holds six ounces.

German Canal Locks.—The locks of the Kaiser Wilhelm Canal in Germany, which is playing an important part in the naval strategy of the present war, are the largest in the world, being 72 ft. longer and 38 ft. wider than the locks of the Panama Canal.—*Machinery.*

NEW DEVICES

THE YOUNG VALVE GEAR

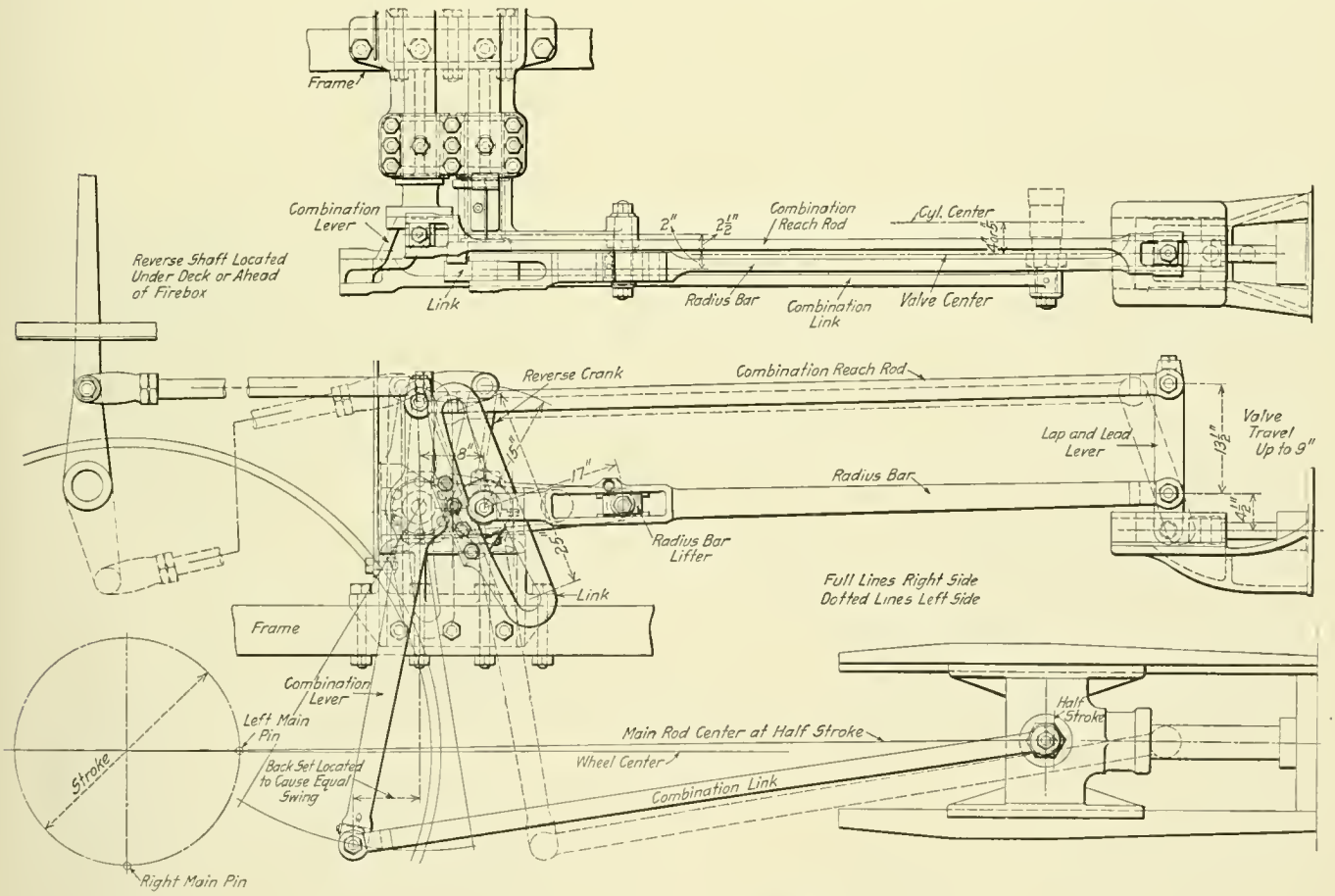
O. W. Young of Schenectady, N. Y., has designed a valve gear for locomotives which has a number of improved features when compared with the Walschaert design. It is actuated entirely by connections to the crossheads and is arranged to give a movement to the valve which materially increases the amount of the port opening for both admission and exhaust, and allows a 14-in. diameter piston valve to handle as large a volume of steam as a 20-in. diameter valve would with the ordinary design of Walschaert gear. It also simplifies the construction, allows easy standardization, and materially reduces the weight of the whole gear.

As will be seen by reference to the illustration, the combination lever which gives the valve its lap and lead movement, the same as in the Walschaert gear, is located some distance back of the crosshead and is keyed to a shaft which extends across the frames. On the opposite end of this shaft the link is keyed and gives the travel to the valve on that side. It will thus be seen that the connections to the crosshead on the right side of the locomotive provide for the lap and lead movement of the valve on the same side and the movement of the link which gives the travel to the valve on the opposite side. The same arrangement applies for the other side. Reversing is effected through a bell crank on either side that moves the radius rod by means of a block in a slotted opening. There are two reach rods, one for each side of the locomotive, and the arrangement is such that when one radius bar is lowered the other is raised. This balances the two gears for reversing, without the use of a counterbalance spring. The radius rod and the connection from the top of the combination lever both join a short link, the end of which is connected directly to the valve stem crosshead.

The link, combination lever and reverse bell crank have a common fulcrum on either side and the bearings for both of the shafts extending across the locomotive are contained in the same casting which is supported by a frame cross-tie.

A number of ellipses have been made with this gear on the valve gear model at the Schenectady plant of the American Locomotive Company. The model was arranged for a valve travel of 8½ in., which was secured with a 42 deg. angle of the link and an 18 in. link block lift. For a 9 in. travel the link angle would be 47 deg. In the case of a Walschaert gear, the link angle is 50 deg. for a 7 in. valve travel with the same amount of block lift and an eccentric circle of 22 in. One setting was arranged to give no lead or pre-admission and the release and closure were very late. At a 25 per cent cutoff the port opening was 15/64 in. This is equal to the opening the Walschaert gear gives with a 3/16 in. lead. If the Walschaert gear is set with no lead, the port opening at a 25 per cent cutoff is only 3/32 in.

In one of the illustrations are shown the ellipses obtained from a Walschaert gear with a 6½ in. valve travel, 1⅛ in. lap and ¼ in. lead, having a sliding block radius bar lifter. On



The Young Locomotive Valve Gear Is Driven Entirely by the Crossheads

gears their tremendous hold upon the railroads of today are low maintenance cost, and the faculty of "staying put"; accessibility for inspection and repairs, and simplicity of the mechanism as a whole and of its parts.

Having adopted a form of valve gear well suited to the exacting requirements of modern power, there should be banished, along with the discarded Stephenson gear, certain venerable illusions regarding valve setting. Squaring an engine has always meant equalizing the cut-off in all positions of the reverse lever, often at the expense of port opening at one end. Almost every road has some engine, or class of engines, which sounds lame even when the valve setting report is perfect. An indicator card from one of these would quite frequently show as high as 20 per cent more work in one end of the cylinder than the other, and yet the cut-off is equal; the engine is perfectly square in the accepted sense.

Very few valve gears are so well designed that both lead and cut-off can be equalized at all points of the cut-off. According to regular practice, or following positive instructions, the valve setter unhesitatingly sacrifices the former for the latter.

The rule should be, square the port opening in running cut-off, say 25 per cent cut-off for passenger service and 50 per cent cut-off for freight service, laying the blame for unequal cut-off on the designer, where it properly belongs. A difference of 2 in. in cut-off will usually be much less apparent, both in the sound of the exhaust and in the indicator card, than a difference of 1/32 in. in port opening. The reason for this should be apparent. At the higher speeds and shorter cut-offs, port opening is simply the lead plus a small increment depending on the ratio of lap and lead to valve travel. At 25 per cent cut-off, in modern engines, port opening amounts to from ¼ in. to ⅜ in. At 70 miles an hour, with the valve open this small amount for a period about equal to the snap shot of a very fast camera shutter, it is almost inconceivable how enough steam can enter the cylinder to produce an indicator diagram. Then think of allowing the opening at one end to exceed by 1/16 in. (which may mean 25 per cent) the amount at the other. Cut-off, on the other hand, is usually a purely imaginary point on a high-speed, or even a moderate-speed locomotive indicator diagram; try to locate it, and then let some one else try. The difference between the two estimates will probably greatly exceed an amount equivalent to 2 in. on the stroke.

Steam engine slide valve motion, while it will always require more or less ingenuity in its detailed application, is in principle beautifully simple. Whatever its type, it consists of a mechanism to combine two movements, one at right angles to the other (i. e.), one must cease when the other is at its maximum and to transmit this combined movement to the valve. There are dozens of ways of accomplishing this, most of which have been tried at one time or another. Quick opening and quick closing of the ports has been the goal toward which most inventors have striven, mostly in vain. Slight gains in this respect over "straight" movement, such as the Stephenson and Walschaert, are found in some modern gears, the gain being largely at one end of the travel; but it may be accepted as a truism that any gear accomplishing this desirable feature to any appreciable extent, as measured by the coal pile, will be too expensive and too complicated for this generation, at least.

Many valve gears, highly suitable to certain special types of locomotives, have been condemned in this country because of mistakes in their early applications. Errors have been most frequently made in proportioning parts to the work to be done. Now and then new valve gears have been tried by designers with an adequate knowledge of some peculiarity of the type, as in the first applications of the Walschaert. Those responsible for these earlier designs failed to take into account the fact that, by giving the Walschaert gear a constant lead equal to the lead of the Stephenson at running cut-off, they greatly shortened the full gear cut-off, to the detriment of the starting power of the engine. This lack of starting power was for a long time believed to be an inherent defect of the valve gear, whereas a little less lap would have overcome the difficulty.

By some builders, too much attention has been paid to theoretical considerations in designing the Walschaert gear. For instance, the location of the reverse shaft to give the best equalization of cut-off in both forward and back motion should be in front of the link, with the lifting arm extending backward. If the reach rod is then connected directly to the upper arm, the motion is forward when the block is at the top of the link. This is contrary to good practice, as in case of the failure of the reach rod, or the slipping out of the reverse lever latch, the engine might be suddenly reversed while running at high speed. In this case, as usual, the simplest arrangement is best; locate the reverse shaft convenient to the reach rod, and arrange the gear so that the slip in forward motion is minimized. In road engines, let the back motion take care of itself.

The bogey of box play, which in certain types of radial gear causes a slight distortion of the valve events, has prevented the application (except in a few special cases, where it seems to be giving entire satisfaction) of the well-known Joy motion. The same argument has been used against one of the most successful valve gears of the day. This is but a technicality. It is extremely doubtful if the momentary shortening of the cut-off at one end, compensated for by a corresponding lengthening at the other, has the slightest effect on the consumption of fuel or the drawbar pull of the locomotive.

As mentioned before, the most common error made in introducing valve gears which are used abroad is that we copy European practice in proportioning the parts, along with the idea. It should not be forgotten that American locomotives are expected to run between shoppings, which means from one to three years with only the most cursory repairs, while European locomotives receive much closer attention. Rubbing surfaces, pin bearings, etc., should be nearly double in area as compared with the best European practice, and renewable bushings used in all cases instead of take-up bearings.

POWDERED FUEL

The following is taken from Bulletin No. 8, entitled "Some Engineering Phases of Pittsburgh's Smoke Problem," issued by the University of Pittsburgh:

The main advantages in the use of powdered fuel may be summed up as follows:

1. Complete combustion and total absence of smoke, when this process is carried out in a properly designed and operated furnace.

2. Losses due to excess air and cooling of furnaces by opening of fire doors are reduced to a minimum.

3. Use of a cheaper grade of bituminous coal, as impurities have very little effect on the successful operation of the process.

4. The ability to meet sudden changes in load, and reducing to a minimum the labor inherent to firing.

Among the disadvantages are:

1. Danger inherent to the storage of large quantities of powdered fuel, giving rise in most cities to the enactment of laws prohibiting the storage of large quantities of this fuel.

2. Inability to secure, at a moderate cost, a satisfactory material to withstand the intense heat developed when operating this type of furnace properly.

3. Tendency of the stronger drafts to carry the fuel through the furnace unburned.

The application of this process to the steam boiler has no doubt largely been hampered by the fact that the maintenance cost in daily operation is high, due to rapid deterioration of brick work. The reliability of all devices as yet applied to boilers, is questionable. It is claimed that savings of 40 per cent have been made when powdered coal was applied to metallurgical processes, such as puddling, heating and reheating furnaces, and that smokeless operation was obtained in all cases.